

HI 2022: INTRODUCTION TO PYTHON FOR HEALTH INFORMATICS

Term Project (Final Project)

Fall 2025

This document describes our HI 2022's term project. For HI 2022's term project, you will tackle common steps in predictive (supervised) machine learning model(s) using a health-related dataset.

Step (1): Finding a publicly available dataset in any healthcare setting

In this step, you need to find and grab a publicly available dataset in any healthcare setting, such as cancer, orthopedic, and cardiovascular. “[Kaggle](#)” could be a good source to find and grab a relevant dataset. In your dataset selection, please select a dataset that suits “supervised” and “predictive” machine learning models. That being said, the dataset should have a list a variable (x values) and an outcome (y value).

Examples of those datasets include, but not limited to:

[1] <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>

(Outcome: diagnosis [M: malignant AND B: benign])

[2] <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

(Outcome: target [0: no disease AND 1: disease])

Step (2): Loading the dataset into your Python IDE

Step (2) involves loading, reading and exploring the dataset. You may need to save the dataset in either your google drive or your local machine (e.g., your PC). Your data exploration may include understanding your dataset size, dataset shape, eliminating irrelevant attributes/features, etc.

Step (3): Data Visualization

In step (3), you need to draw three to four types of data visualization to better understand the dataset. Your data visualization may include bar chart, scatter chart, pie chart, etc.

Step (4): Making the dataset available for “Supervised” Machine Learning algorithms

In this step, you need to select your variables (x values) and the outcome (y value). You may also need to do data encoding (turning all data into numeric values), and for sure data normalization. Furthermore, you also need to split your data into train/test sets using K-fold cross validation.

Step (5): Building and evaluating machine learning predictive models

In this step, you will build, train, and test two predictive models, using **Decision Tree** and **Naïve Bayes**. The evaluation metrics should include -at least- **Accuracy**, **Precision**, and **Recall** for every target class.

Step (6): Comparing the predictive models you built using Decision Tree and Naïve Bayes

You need to compare the accuracy performance of your Decision Tree model with the Naïve Bayes model.

Step (7): Draw conclusion

This step should briefly conclude the term project and discuss its current limitations and shortcomings.

Implementation, Code, Documentation, deliverables, and Teamwork

- **Mandatory [implementation]:** You will create a GitHub repo, and put your implementation and documentation there. You please add me to your GitHub repo as a collaborator. My GitHub ID is “aptafti”.
- **Mandatory [deliverables]:** You will need to upload:
 - 1) .ipynb file
 - 2) Dataset (e.g., .csv file)
 - 3) GitHub ReadMe.

Rubric

Approx. % of Grade	Excellent (100%)	Adequate (80%)	Poor (60%)	Not Met (0%)
15% Design and Implementation	No errors, it works correctly and meets the specifications.	Minor details of the term project specification are violated, project functions incorrectly for some inputs/cases.	Significant details of the specification are violated, project often exhibits incorrect behavior/functionality.	Project only functions correctly in extremely limited cases or not at all.
5% GitHub ReadMe	Well-organized ReadMe, and accurate performance analysis.	Minor issues in ReadMe or performance analysis.	Significant issues in ReadMe or performance analysis.	No GitHub ReadMe.